

# NightWatch

Hostel night attendance for BITS Pilani, Dubai Campus

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## Video script, pitch, demo and judges' questions

IEEE BPDC CampusOps Hackathon · Department Challenge Track (Hostel Administration)

Problem statement PS04: Night Attendance Scanning

Deliverables: demo video (3 minutes or less) · public GitHub repository · finalist Q&A

Code: [github.com/Anirudh1365/nightwatch](https://github.com/Anirudh1365/nightwatch)

Team: \_\_\_\_\_

Who presents what: \_\_\_\_\_

How to use this document: Part 1 is the script for the 3-minute video, shot by shot. Part 2 is the live pitch (about 5 minutes) if you are asked to present. Part 3 is the live demo, step by step, with a setup checklist and a fallback for every step. Part 4 prepares you for the finalist Q&A, grouped under the four topics the organisers listed, including cost. The appendix has logins and commands. Shaded blocks are lines to say out loud; say them in your own words.

## Part 1 · The 3-minute video

The organisers ask for a video of **3 minutes or less** that shows the problem and the solution, a working demonstration, the users and how they use it, and the technical choices and their impact. This script covers all four in about **2 minutes 50 seconds**, so you have a little room.

The narration is about 390 words. At a normal speaking pace that is just under 3 minutes, so don't add lines. If the cut runs long, shorten the demo clips, not the words.

### Before you record

1. In nightwatch/backend run **python run\_demo.py --pretend 23:35 --reset**. This gives fresh demo data and about 25 minutes of on-time check-ins.
2. Log everyone in beforehand (see Part 3, setup checklist), so the certificate warning never appears on camera.
3. Turn on Do Not Disturb on the laptop and both phones. Close other tabs. Brightness up.
4. Record each segment as its own clip. You can retake one clip without redoing the rest.

### How to record and edit (free tools)

- **Laptop screen:** press **Win + Alt + R** to record the current window with Xbox Game Bar (Win + G opens it), or use OBS Studio for the full screen.
- **Phone screen:** use the phone's own screen recording (iPhone: Control Centre; Android: Quick Settings).
- **The scan itself:** film it with a second phone, so viewers see the phone pointed at the reception screen. That shot shows the idea best.
- **Voice:** record the narration separately with earphones and a mic in a quiet room. It sounds better than talking while clicking.
- **Edit:** Clipchamp comes with Windows 11. Put the clips on the timeline, cut out waiting and typing, lay the narration on top and turn on auto-captions.
- **Export** 1080p MP4. Check the length is under **3:00**. Upload it as unlisted on YouTube (or as a Google Drive link anyone can view) and add the link to the README.

### The script

#### 0:00 The problem (20 s)

##### On screen:

- Title card: **NightWatch: night attendance for the BITS Pilani Dubai hostels.**
- Then simple text slides: "Paper list, room to room", "Late? Write your room in a notebook", "Gate passes = WhatsApp screenshots".

##### Say:

Every night in our hostels, an attendance taker walks room to room with a paper list. Students who come back late write a room number in a notebook at reception. Nobody checks the name or the time. Gate passes are screenshots in a staff WhatsApp group, and wardens hear about absences the next morning.

#### 0:20 The solution and who uses it (20 s)

##### On screen:

- One slide with the five users and one line each: student checks in, guard runs the reception screen and ID desk, taker does a short round, warden sees exceptions, parent approves passes by email.

Say:

NightWatch replaces that with a web app. Students check in at their block's reception with their phone or their university ID card. The attendance taker only visits the rooms that didn't check in. Parents approve gate passes from an email link. And wardens see the exceptions the same night.

### 0:40 Working demo: phone check-in (20 s)

On screen:

- Reception screen full screen, the bar under the QR emptying and the code changing.
- Second-phone shot: Aarav's phone scanning the screen, then the green **Marked present**.

Say:

This is the screen at Block A's reception. The code changes every fifteen seconds. Aarav is on the campus Wi-Fi, so he scans it and he's checked in, in about ten seconds. Check-in is open from ten to ten-fifteen for early sleepers, and from eleven-thirty to midnight for everyone else.

### 1:00 Working demo: someone off campus is refused (15 s)

On screen:

- Vivaan's phone (on the laptop hotspot) scanning the same screen, then the red **Connect to the campus Wi-Fi** message.

Say:

Vivaan scans the same code, but his phone isn't on the campus Wi-Fi. The server checks which network the request came from, not GPS, so it refuses him. A forwarded screenshot stops working within thirty seconds anyway.

### 1:15 Working demo: ID card at the desk (15 s)

On screen:

- ID desk page. The guard scans Kabir's card, or types **2023A7PS0004U** and presses Check in. Kabir, room A102, present.

Say:

No phone? The guard scans the student's university ID card at the desk. Same rules, and the guard sees the name and room on screen.

### 1:30 Working demo: gate pass (20 s)

On screen:

- Ishaan's phone: request a pass. Terminal: the parent's email. Parent page: **Approve**. Warden page: **Approve**. Ishaan's phone: **Approved**.

Say:

Ishaan asks for a gate pass. His parent approves it from a one-time email link, then the warden approves. The guard sees it, and Ishaan is excused tonight. No more WhatsApp screenshots.

### 1:50 Working demo: the taker's round (15 s)

On screen:

- Taker page: **Rooms to visit**. Pick **Asleep** and tap Present for one student; tap Absent for another.

Say:

The taker's list only has students who haven't checked in, plus five random spot checks. Marking someone present by hand needs a reason, and an empty room alerts the warden straight away.

## 2:05 Working demo: the warden (20 s)

### On screen:

- Warden page: **Absent tonight, Tonight's register, Change history** showing Vivaan's refused try and the taker's marks, click **Download CSV**, open **Last 14 nights**, click **Run the 14-night check**.

### Say:

The warden sees tonight's exceptions live: absent, late, not checked in, and marked by hand. Every change goes into a history nobody can edit, with who did it and when. Reports download as spreadsheets, and a student with three absences in two weeks is flagged by email.

## 2:25 Technical choices and impact (30 s)

### On screen:

- The architecture diagram from the README's Technical overview, then a few seconds of the code.
- End card: **NightWatch** + the GitHub link.

### Say:

Under the hood, it's a Python FastAPI server with a MySQL database, and plain web pages, so there's no app to install. QR codes are signed and expire, passwords are hashed, and every page checks the user's role.

The result: a ten-second check-in instead of a paper round, a much shorter round for the taker, and real numbers for the wardens, including for the girls' curfew pilot. Thanks for watching.

Timings assume about 2.5 spoken words a second. Time your own recording; if a segment runs over, cut the clip under it, not the words.

## Part 2 · The live pitch (about 5 minutes)

Use this only if the finalists are asked to present. If the round is Q&A only, go straight to Part 4.

### The one-line version

NightWatch replaces the 10:30 PM paper roll call with a 10-second check-in at the block reception, by phone or university ID card, so the attendance taker only visits the rooms that actually need a visit, and the wardens get the same night's absences and a real curfew report instead of a notebook.

### 0:00 The problem today

Every night an attendance taker walks room to room with a paper list. Students who come back late write their room number in a notebook at reception. Nobody checks their name or the time.

Gate passes work like this: a parent emails the warden, the warden posts a screenshot in a staff WhatsApp group, and the guard scrolls through it. The pass is valid for one day only.

And right now, management is running a one-month pilot with an 11:00 PM curfew for girls. They said they'll reverse it if late coming continues, but today nobody has the numbers to show whether it's working.

### 0:45 What NightWatch does

The problem statement asks us to record night attendance by scanning a university ID or another approved credential at set locations. NightWatch does that in three parts. Students check themselves in at their block's reception, with their phone or their ID card. The taker only visits the rooms that didn't check in, plus a few random spot checks. And gate passes go from parent to warden to guard without WhatsApp screenshots.

It runs in any phone browser. There's no app to install.

### 1:15 How check-in works

There are two check-in times: 10:00 to 10:15 PM for students who sleep early, and 11:30 to midnight for everyone else. The student checks in once, in either, by scanning a QR code on the screen at their own block's reception. That code changes every 15 seconds, so a screenshot sent to a friend stops working almost immediately.

The phone also has to be on the campus Wi-Fi. The server checks the connection itself, so a fake-GPS app doesn't help, and a student who's out on mobile data can't check in.

No phone, or a dead battery? The guard at reception scans the student's university ID card: the barcode with any camera, or a tap on a USB card reader. Same time rules, and still only one check-in a night.

After midnight they can still check in, but it's marked late. That late data is exactly what the curfew review needs.

### 2:00 The taker's round

Instead of knocking on every door, the taker gets a short list: only the students who didn't check in, plus about five random students who did. The random ones keep everyone honest, because nobody knows who'll be checked.

If the taker marks someone present by hand, they must choose a reason: asleep, sick, phone problem or other. If a spot-checked room is empty, the warden gets a high-priority alert that same night.

## 2:30 Gate passes

The student requests a pass in NightWatch. The parent gets an email with a one-time link and approves it in one tap, with no account needed. Then the warden approves it. The guard sees today's approved passes on their own page, and the student is automatically excused from attendance that day.

## 3:00 What the wardens get

Wardens see tonight's unexcused absences as they happen, not the next morning. The register shows every student and marks the exceptions: late, absent, not checked in, or marked by hand.

Every check-in attempt, including refused ones, every hand mark with its before and after, and every gate-pass decision goes into a change history that nobody can edit or delete. The register, the 14-night table and the history all download as spreadsheet files.

One click emails any student with three absences or five hand-marked check-ins in two weeks and asks them to meet the warden. For the girls' curfew pilot, the 14-night table is the evidence management said they need.

## 3:40 Why it's hard to cheat

| If a student tries to...                                   | What stops it                                                                              |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Send a screenshot of the QR to a friend                    | The code changes every 15 seconds and is accepted for about 30 seconds at most.            |
| Check in from outside campus                               | The server sees the phone is not on campus Wi-Fi and refuses the check-in.                 |
| Fake their location with a GPS app                         | NightWatch doesn't use GPS. The check is done by the server, not the phone.                |
| Check in and then leave, or check in from another building | Random spot checks. An empty room raises a high-priority alert to the warden.              |
| Hand their ID card to a friend                             | The guard at the desk sees the student's name and room on screen. Spot checks still apply. |
| Scan their card at another block's desk                    | Refused: the card belongs to a different block. The refusal is kept in the change history. |
| Always get marked by hand "because my phone died"          | Every manual mark needs a reason; five in 14 nights flags the student to the warden.       |

## 4:10 What it doesn't do (say this, judges respect it)

We want to be clear about the limits. The Wi-Fi check proves a student is on campus, not that they're in their own block. A friend in the lobby could forward the live code to someone elsewhere on campus. The short code life and the random spot checks are our answer to that. If a block needs more, a small Wi-Fi router at its reception is a cheap upgrade.

NightWatch needs the campus network and its server. If either is down, it can't take check-ins and the taker goes back to the paper round. We haven't built an offline mode yet. And parent approval relies on the parent's email, which is exactly what the process relies on today.

## 4:35 What's next

Phase 2 has two parts. First, move NightWatch into the ERP as a Hostel section, next to Academics, HR and Faculty, with ERP login and the university's own student list. Only the ERP team can add a section,

so today it runs beside the ERP. Second, connect the main-gate face scanner log to catch anyone who left without a pass. We haven't built either, because we don't have access yet.

#### **4:55 Close**

NightWatch turns a paper round and a notebook into a 10-second check-in, a shorter round for the taker, and real data for the wardens. Let us show you.

## Part 3 · The live demo (about 6 minutes)

### Setup checklist (do this 15 minutes before)

1. Laptop on BITS-Student Wi-Fi. Phone(s) on BITS-Student, battery charged, brightness up.
2. On the laptop: open a terminal in nightwatch/backend and run **python run\_demo.py --pretend 23:35 --reset**. This loads fresh demo data and makes the server act as if it's 23:35, so check-in is open whatever the real time is. The pretend clock keeps ticking: you have about 25 minutes of on-time check-ins. If the demo slips, stop the server (Ctrl + C) and run **python run\_demo.py --pretend 23:35** again, without --reset, which keeps the data.
3. Write down the phone address it prints next to "phones on BITS-Student use this" (for example <https://10.30.83.168:8000>). It changes if the laptop reconnects to Wi-Fi.
4. Turn on the laptop's Mobile hotspot (Windows settings). It's only used for the refusal in step 3.
5. If Windows Firewall asks about Python, click Allow (tick public networks too).
6. Laptop, Chrome normal window: go to <https://localhost:8000>, log in as **guard.a@demo.edu**, open the reception screen and press Full screen. This is "the reception display". In a second tab of the same window, open the guard page and click **Open ID desk**.
7. Laptop, Chrome private (incognito) window: log in as **warden.a@demo.edu**. Laptop, Edge: log in as **taker.a@demo.edu**. (Each browser window type holds one login.)
8. Phone: open the phone address, tap Advanced → Proceed on the certificate warning, log in as **aarav@demo.edu**. Do this before the judges arrive so they don't see the warning.
9. Keep the terminal visible or one Alt+Tab away. Every check-in prints a [check-in] line there, and the parent's email appears there too.

All passwords: demo123. If the laptop's IP changes, the phone shows the certificate warning once more; tap Proceed again.

### Step 1 The reception screen Laptop (reception display)

#### Do:

- Show the full-screen Block A QR code. Point at the bar under the code emptying.

#### Say:

This is the screen at Block A's reception. The code changes every 15 seconds, so a photo of it is useless half a minute later.

**Expect:** The code visibly changes when the bar runs out.

**If it fails:** If it says "Can't reach the server", check the terminal is still running and the laptop is still on Wi-Fi. It reconnects by itself within 3 seconds.

### Step 2 A student checks in Phone (Aarav, on BITS-Student)

#### Do:

- Show the student page: "Not checked in", with both check-in times (22:00–22:15 or 23:30–00:00).
- Tap **Scan QR at reception**, point the phone at the laptop screen.

#### Say:

Aarav is on the campus Wi-Fi. He scans the code at his block's reception, and he's done. That took about ten seconds.

**Expect:** Green "Marked present". The terminal prints a [check-in] line: Aarav, his phone's address, present.

**If it fails:** If the camera doesn't open, check Safari/Chrome camera permission for the site. If the scan says "QR code expired", just scan again (the code had just changed).

### Step 3 Someone not on campus Wi-Fi is refused Phone on the laptop hotspot (Vivaan)

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**Do:**

- Switch the phone's Wi-Fi to the laptop's hotspot. Open <https://192.168.137.1:8000> (Proceed on the warning if shown), log in as **vivaan@demo.edu**.
- Scan the same reception screen.

**Say:**

Now Vivaan's phone is on a network that isn't the campus Wi-Fi. In real life that's mobile data. For the demo we use the laptop's hotspot, because a phone on mobile data can't reach a laptop on campus at all. Same code, same block, and the server refuses him.

**Expect:** Red "Connect to the campus Wi-Fi (BITS-Student), not mobile data". The terminal prints REFUSED, not on campus Wi-Fi.

**If it fails:** A second phone makes this step faster: keep phone 2 on the hotspot from the start. Afterwards, switch phone 1 back to BITS-Student.

### Step 4 ID card at the desk Laptop (Chrome, ID desk tab)

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**Do:**

- Switch to the ID desk tab. In **Card number**, type **2023A7PS0004U** (Kabir's ID number) and press **Check in**. A USB tap reader does exactly this: it types the number and presses Enter.
- If you have a printed barcode of that number, use **Scan barcode with camera** instead.
- Optional: type **2023A7PS0011U** (Ananya, Block G) to show a card from another block being refused.

**Say:**

Kabir's phone is dead. He gives his university ID card to the guard. The tap reader or the camera reads the number, and he's checked in with the same rules as the phone. The guard sees his name and room, so a borrowed card is easy to spot.

**Expect:** Kabir, Room A102, present, at the top of Recent scans. The terminal prints a [check-in] line.

**If it fails:** If it says check-in is closed, the pretend clock has run past midnight: restart with **python run\_demo.py --pretend 23:35** (no --reset).

### Step 5 Gate pass: student, parent, warden Phone (student), terminal, laptop (warden)

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**Do:**

- On the phone (back on BITS-Student), log out and log in as **ishaan@demo.edu**. Under Request a gate pass, keep today's date, type a reason, tap **Send request**.
- In the terminal, show the email to the parent. Copy the <https://.../parent.html?t=...> link and open it (laptop or phone). Tap **Approve**.
- In the warden window, press Refresh: the pass is under "Gate passes waiting for you". Click **Approve**.
- On Ishaan's phone, reload: My passes shows Approved.

**Say:**

Ishaan asks for a pass. His parent gets a one-time link by email and approves with one tap, with no account needed. Then the warden approves. The guard sees it on his page, and Ishaan is excused from tonight's attendance. No WhatsApp screenshots.

**Expect:** Parent page says "Thank you. You approved the request." Opening the same link again says it's already used.

**If it fails:** In the demo, emails are printed in the terminal instead of being sent. Say so if asked: switching to real email needs the university's mail server settings.

## Step 6 The taker's round Laptop (Edge, taker)

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**Do:**

- Show "Rooms to visit": only students who haven't checked in. Ishaan isn't on the list, because he's excused.
- For one student, pick **Asleep** and tap **Present**. For another, tap **Absent** and confirm.

**Say:**

The taker doesn't walk every corridor. They see only who's missing. Marking present by hand needs a reason, so we can spot the student whose phone is always "dead". Marking absent alerts the warden right away.

**Expect:** The marked students disappear from the list. Tonight's counts update.

**If it fails:** Spot checks appear once students have checked in. Aarav or Kabir may show up as a spot check. That's fine, and worth pointing out.

## Step 7 The warden's view Laptop (private window, warden)

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**Do:**

- Press Refresh. "Absent tonight, no gate pass" shows the student the taker just marked absent.
- Open **Tonight's register**: every student, with the exceptions marked (late, absent, not checked in, marked by hand).
- Open **Change history**: Vivaan's refused try, Kabir's ID card check-in, the taker's marks (before → after) and the gate pass decision, each with who and when. Click **Download tonight** and open the CSV.
- Open **Last 14 nights**: one row per night with on time, late, absent and marked by taker.
- Click **Run the 14-night check**. Rohan is flagged; show his email in the terminal.

**Say:**

The warden sees tonight's absences tonight. Every change is in a history nobody can edit: who did what, when, and what it was before. Everything downloads as a spreadsheet. This table is two weeks of on time, late and absent, the evidence management needs for the curfew pilot. And one click emails every student who crossed the limit.

**Expect:** Rohan, Room A105, flagged with his absences; an email line in the terminal.

**If it fails:** The 14-night table uses demo history created by --reset. Say so if asked: "these past nights are sample data; tonight is live."

Close the demo: "That's a full night: check-in, a refusal, an ID card at the desk, a gate pass, the taker's round and the warden's report. Happy to take questions."

## Part 4 · Finalist Q&A

The organisers said finalists should be ready to discuss four things: problem understanding; solution and technical choices; integration and usability; and cost, limitations and impact. The questions below are grouped the same way. Keep answers short and honest. If you don't know something, say what you'd need to find out.

### 1. Problem understanding

**Q: What is actually wrong with the current process?**

A: Three things. It's slow: one person walks every room every night. It's unverified: late students write a room number in a notebook, with no name or time checked. And it leaves no usable record: gate passes are WhatsApp screenshots, and wardens learn about absences the next morning. Management can't tell whether the girls' curfew pilot is working because nobody has the numbers.

**Q: Who are the users?**

A: Students (check in, request passes), guards (run the reception screen and the ID desk, check passes), attendance takers (short round), wardens and the chief warden (exceptions, approvals, reports) and parents (approve passes from an email link, no account).

**Q: Why did you leave out complaints?**

A: The university's ERP support portal already handles complaints, and the Student Council asked students to use it. We didn't want to build a second channel.

### 2. Solution and technical choices

**Q: What if a student sends the QR code to a friend who isn't in the hostel?**

A: The friend must be on campus Wi-Fi and use it within about 30 seconds. If they're off campus, it's refused. If they're elsewhere on campus, it can work, which is why spot checks exist: that student's room may be checked, and an empty room is a high-priority alert.

**Q: Why not use GPS?**

A: GPS is easy to fake with free apps and unreliable indoors. The Wi-Fi check happens on the server, and the phone can't lie about which network its connection came from.

**Q: Why not only use the ID card, since the brief mentions it?**

A: We do use it, at the reception desk. But one desk per block means everyone queues in the same half hour. Phones let the whole block check in at once, and the desk handles anyone without a working phone. Both go through the same rules.

**Q: What if a student lends their ID card to a friend?**

A: The guard is at the desk and sees the name and room on screen when the card is scanned. Random spot checks still apply to ID card check-ins, the same as phone ones.

**Q: How do you know the records haven't been changed?**

A: Every check-in attempt, hand mark, gate pass decision and flag email adds a row to the change history, with who, when, and before → after. The app has no way to edit or delete those rows. Someone with direct access to the database could still change it, so in production only IT should have that access, and the database is backed up every night.

**Q: How are accounts and passwords protected?**

A: Passwords are stored hashed (bcrypt), logins use signed tokens that expire after 12 hours, every page checks the user's role and block, and parent links work once and expire after 24 hours.

**Q: Why plain web pages and not an app?**

A: Nobody has to install anything, it works on every phone, and the university's IT team can run it like any website. The QR scanner and fonts are stored with the app, so it doesn't depend on outside websites.

**Q: Does this track students' location?**

A: No. It stores the time of the check-in and the network address of that one request, nothing else. No GPS, no background tracking.

### **3. Integration and usability**

**Q: Why isn't it inside the ERP, like Academics, HR and Faculty?**

A: That's where it should end up, as phase 2. Only the university's ERP team can add a section to the ERP, and we needed something we could build and show working for this round. We've already matched the ERP's look so it feels familiar. Moving in means students log in with their ERP account and the student and room list comes from the university instead of our own copy.

**Q: Would the Wi-Fi check still work from inside the ERP?**

A: Mostly. The ERP runs on Oracle Cloud, outside campus. From there, every phone on campus Wi-Fi shows up with the campus's shared internet address, not its own campus address. That still tells campus Wi-Fi apart from mobile data, which is the main check. We'd lose the detail of which part of the campus network the phone is on, and the campus address list would need updating if it ever changes. The rotating QR at the reception still proves the student is at their block. The other option is to keep the check-in step on a small campus server and put everything else in the ERP.

**Q: Does it work with the ID cards students already have?**

A: Yes, without changing the cards. The camera reads the barcode, which we expect holds the student's ID number (we'll confirm with a real card). The chip can be read by a USB tap reader, like the one on the printer top-up machine. We don't know which chip the card uses or how its number maps to students, so the first time an unknown card is tapped, the warden links it to the student once. We haven't tested a real card yet, and we won't buy a reader until we know the chip type.

**Q: What about the main-gate face scanner?**

A: Also phase 2. Its exit and entry log would show anyone who left without a pass. We didn't integrate it because we have no access to its data.

**Q: What would the university need to give you to go live?**

A: A small server on the campus network from IT, the campus Wi-Fi address ranges, the student list with rooms and parent emails (a one-off export from the ERP), the mail server settings for sending emails, and a screen at each block's reception.

**Q: How hard is it for staff to learn?**

A: There's one login page for everyone, and each person lands on their own page with only what they need. The guard opens the reception screen once and leaves it on. The taker gets a list with Present and Absent buttons. It looks like the ERP they already use.

### **4. Cost, limitations and impact**

**Q: What does it cost?**

A: No licence fees: everything we used is free, open-source software, and students use their own phones. The hardware is one small server and a screen at each of the six receptions, and old monitors are enough. It costs close to nothing if IT reuses equipment, and roughly AED 8,000 one-off if everything is bought new. The table below has the details.

| Item                      | What's needed                                                    | One-off cost (estimate)                                                                                             |
|---------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Server                    | 1 small server on the campus network (2 CPU, 4 GB RAM is plenty) | AED 0 as a virtual machine on IT's existing servers; about AED 1,500–2,500 for a new mini PC                        |
| Reception screens         | 6, one per block                                                 | AED 0 with an old monitor and PC; about AED 400–700 each for a basic Android tablet (about AED 2,400–4,200 for six) |
| ID desk                   | A camera to read the barcode                                     | AED 0: the same tablet or laptop camera                                                                             |
| USB tap reader (optional) | 1 per block desk                                                 | About AED 50–250 each, only once the card's chip type is known                                                      |
| Software                  | Python, FastAPI, MySQL Community, open-source libraries          | AED 0, no licence fees                                                                                              |
| Email                     | The university's existing mail server                            | AED 0                                                                                                               |
| Staff time                | IT sets it up and loads the student list each semester           | About 1–2 days to set up, then a few hours each semester                                                            |

Prices are rough UAE retail estimates (October 2026), not quotes. Running costs are close to zero: electricity and IT's existing backups.

**Q: What are the limitations?**

A: Say these plainly. (1) The Wi-Fi check proves "on campus", not "in this block"; the QR and spot checks cover that. (2) A borrowed ID card relies on the guard looking at the screen. (3) It needs the campus network and the server; if they are down, it's back to the paper round, because there's no offline mode yet. (4) Parent approval trusts the parent's email, the same as today. (5) We haven't tested a real BITS ID card or load-tested all six blocks at once. (6) In the demo, emails are printed instead of sent, and the past 14 nights are sample data.

**Q: What if the Wi-Fi or the server goes down?**

A: Then NightWatch can't take check-ins and the taker does the full round on paper, which is exactly what happens today. NightWatch removes work when it's up; it doesn't add risk when it's down. An offline mode for the taker's phone is on our list, but it isn't built.

**Q: What if a student's phone is dead or they don't have one?**

A: They check in with their ID card at the reception desk. If that's not possible either, the taker marks them present by hand with the reason "phone issue". Five manual marks in 14 nights flags it to the warden.

**Q: What's the impact?**

A: The taker visits only the students who didn't check in, plus five spot checks, instead of every room. For example, if 90 of 100 students in a block check in themselves, the round drops from 100 doors to about 15. Wardens learn about absences the same night instead of the next morning. Late returns are recorded with a name and time instead of a room number in a notebook. Gate passes leave a record instead of screenshots. And management finally has the late-return numbers for the curfew pilot.

**Q: Will it handle every student checking in at once?**

A: Each check-in is one short request, so even every student in all six blocks checking in within half an hour is light work for one small server. We haven't load-tested it at that size yet; that would be part of a pilot in one block.

**Q: Is the 14-night data real?**

A: The past nights in the demo are sample data we generated. Tonight's check-ins in the demo are live.

## Appendix · Logins and commands

### Demo logins (password for all: demo123)

| Role                           | Email                                                                                 |
|--------------------------------|---------------------------------------------------------------------------------------|
| Chief warden (all blocks)      | chief@demo.edu                                                                        |
| Block A warden / taker / guard | warden.a@demo.edu · taker.a@demo.edu · guard.a@demo.edu                               |
| Block G warden / taker / guard | warden.g@demo.edu · taker.g@demo.edu · guard.g@demo.edu                               |
| Block A students               | aarav, vivaan, ishaan, kabir, reyansh, arjun, dhruv, vihaan, rohan, karan (@demo.edu) |
| Block G students               | ananya, diya, saanvi, myra, aadhya, kiara, riya, meera, tara, nisha (@demo.edu)       |

One login page for everyone. ID numbers for the ID desk, in the order above: Block A 2023A7PS0001U to 0010U (Kabir is 0004U), Block G 2023A7PS0011U to 0020U.

### Commands (run in nightwatch/backend)

| Command                                    | What it does                                                                                                           |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| python run_demo.py --pretend 23:35 --reset | The demo: fresh data, server clock set to 23:35 (about 25 minutes of on-time check-ins), HTTPS, campus Wi-Fi check on. |
| python run_demo.py --pretend 23:35         | Restart the pretend clock but keep tonight's data.                                                                     |
| python run_demo.py                         | Same server with real time (for a demo actually held at night).                                                        |
| python run_demo.py --no-wifi-check         | Rehearsal off campus: the Wi-Fi check is off.                                                                          |
| python seed.py then python test_flow.py    | Automatic tests (30 checks). Run seed.py again afterwards.                                                             |
| Ctrl + C in the terminal                   | Stops the server.                                                                                                      |

### Addresses

- Laptop: <https://localhost:8000>
- Phones on BITS-Student: the 10.30.x.x address printed by run\_demo.py
- Phones on the laptop hotspot: <https://192.168.137.1:8000> (check-in is refused here, on purpose)

### Facts to have ready

- 6 hostel blocks (4 boys, 2 girls); the demo data has 2 of them (Block A, Block G).
- Check-in 22:00–22:15 or 23:30–00:00, once a night, by phone or ID card; after 00:00 it's allowed but marked late. A "night" runs noon to noon.
- Main gate curfew: 11:30 PM boys; 11:00 PM girls (one-month pilot). Girls' hostel in-time: 12:00 midnight.
- Flag rule: 3 absences or 5 manual marks in any 14 nights leads to an email and a meeting with the warden.